

The empirical recurrence for OEIS A237335: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A237335 records “Empirical recurrence of order 59”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 506$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237335 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with the maximum minus the minimum of every 2×2 subblock equal.”. It has offset 1 and begins

15082, 821234, 39609294, 2137181038, 114598315886, 6318680517398, 350887832861914, 19674122669454962, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:06:09, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 506$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 506$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1012$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	96	c_{16}	54511939490975873301	c_{31}	82516013154375959175260463552
c_2	−768	c_{17}	−323994533015450689437	c_{32}	587961512709604226591444608
c_3	−144583	c_{18}	−1896945537002544508359	c_{33}	−434597259495358647173627778304
c_4	2829202	c_{19}	13079140648558270326783	c_{34}	46416367635108081332516404480
c_5	76621036	c_{20}	46839605666590539966105	c_{35}	1775212858128677213203483541504
c_6	−2048554707	c_{21}	−371664639002531175405612	c_{36}	−318006915646874448230062952448
c_7	−17019483384	c_{22}	−830972909954014455908246	c_{37}	−5626288700514175098561773299712
c_8	683567240153	c_{23}	7681265156740366030560244	c_{38}	1255732865667653650897975547904
c_9	1086744525507	c_{24}	10619637168892207960568400	c_{39}	13807215782241199385055922049024
c_{10}	−127743080457938	c_{25}	−118011843089118668653251200	c_{40}	−3442234102847103154187638046720
c_{11}	217443164912333	c_{26}	−96626201581965824262347184	c_{41}	−26109888585139089543294480564224
c_{12}	14645443896234312	c_{27}	1368693452985749224973135952	c_{42}	6937000551716746743500393627648
c_{13}	−53088580372811046	c_{28}	600052081413121464819195152	c_{43}	37739819299829870946346945806336
c_{14}	−1088187773179562294	c_{29}	−12116333080185364478120896704	c_{44}	−10522321819436677877472574570496
c_{15}	5355964651067399703	c_{30}	−2186663074827188730540497504	c_{45}	−41184702458694573813266893766656

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A237335.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.